

Practice Problem Set -02 (Leibnitz's theorem)

Q.01. if $y = (\sin^{-1}x)^2$, show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$. Hence find $(y_n)_0$.

Q.2. if $y = e^{a \sin^{-1}x}$, prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + a^2)y_n = 0$ Hence find the value of y_n at $x = 0$.

Q. 3. if $y = \sin^{-1}x$, prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$. Also find y_n .

Q. 4. if $y = \tan^{-1}x$ prove that $(1 - x^2)y_{n+2} + 2nxy_n + n(n - 1)y_{n-1} = 0$. Also find $y_{n(0)}$.

Q. 5. if $y = \cos(m \sin^{-1}x)$, prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} + (m^2 - n^2)y_n = 0$

Q. 6. if $y = \sin(m \sin^{-1}x)$, prove that $(1 - x^2)y_2 - xy_1 + m^2y = 0$ and

$(1 - x^2)y_{n+2} - 2(n + 1)xy_{n+1} + (m^2 - n^2)y_n = 0$ also find $(y_n)_0$.

Q. 7. if $y = e^{m \cos^{-1}x}$, prove that (i) $(1 - x^2)y_2 - xy_1 = m^2y$.

(ii) $(1 - x^2)y_{n+2} - 2(n + 1)xy_{n+1} - (m^2 + n^2)y_n = 0$. Also find $(y_n)_0$.